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Obtaining Semi-Formal Models from Qualitative Data: From Interviews Into BPMN Models in User-Centered Design Processes

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ABSTRACT

Gathering qualitative user data in a user-centered design process is one of the very early steps to create interactive systems. However, generating structured models from qualitative data towards descriptions that can be used for the implementation of interactive systems and prototypes raises various challenges, such as a strong influence of the modeler's knowledge and their interpretation of the gathered qualitative data. Introducing the modeler's bias may result in a system implementation which does not fully represent the information provided in the original qualitative data, generating an unwanted gap between what the user needs and what the system provides.

To address this challenge, in this paper we present a structured and manual transformation method, which enables a modeler to create BPMN models from interview data by reducing the modeler's individual influence on the resulting BPMN model. We evaluate this approach in the context of the implementation of persuasive systems, which should support changing unwanted work-related habits. Therefore, we conducted unstructured interviews with office workers, thinking aloud interviews, in which we asked office workers to imagine a situation where they showed an unwanted work-related habit and to describe this habit together with an alternative behavior. In a quantitative experimental study, we then asked study participants to create BPMN models either with or without our new transformation method.

Our analyses showed that when using our method, different participants created very similar BPMN models of the habits, even with little training. We conclude that the major contribution of our work is that the presented method can be applied to the creation of structured models from unstructured interview data. This method that makes use of rich interview data is suitable for the design and implementation of interactive systems.

1. Introduction

The design and engineering of interactive systems faces the challenge of not only creating good software but considering the specific needs and requirements of the addressed user group (Preece et al., 2015). Norman (2005) argues that users very unlikely adapt to a technical system but rather the system needs to fit the user's needs and desires. In particular, this applies to the development of persuasive systems as they are "computerized software or information systems designed to reinforce, change or shape attitudes or behaviors or both without using coercion or deception" (Oinas-Kukkonen & Harjumaa, 2008, p. 164) and thereby not only need to address users' needs but to affect users' behavior persistently. Examples of such systems are, for instance, fitness tracker devices, which remind the user to move or exercise frequently (Konstanti & Karapanos, 2018; Wang et al., 2018), a system which supports workers in an office to follow rules due to sustainability (Lockton et al., 2014), or a system aiming at work-related habits (Langel et al., 2018). Research has shown that persuasive systems profit from a high level of individualization (Fogg, 2009; Oinas-Kukkonen & Harjumaa, 2009), which further strengthens the need for considering the user and the user's individual behavior in the development process of a persuasive system.

To address this challenge, the human-computer interaction community developed and investigated various design methodologies and paradigms. One design paradigm, which has been successfully applied in the past is referred to as *user-centered design* (Abras et al., 2004). User-centered design includes various steps in an iterative fashion including various types of data gathering approaches (using qualitative and quantitative methods) as well as the use of prototypes for evaluation designs that include the user in each step (Abras et al., 2004). With this work, we aim at developing a method as part of a user-centered design process that addresses the development of persuasive systems with the specific focus on an early phase of the design process, mainly relying on qualitative data such as gathered from interviews with the user (Preece et al., 2015). This approach has been mainly motivated by a project focusing on changing work-related habitual behavior (habits) (Langel et al., 2018) where habits can be defined as

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result of a process by which a stimulus generates an automatic response based on learned stimulus-response associations (Gardner, 2015). Addressing habits raises the need for a high level of individualization of the interactive system to support habit change in behavior (Sonnentag et al., 2021) that distinct this work from existing results from health-related habit systems (Orji & Moffatt, 2018) or sustainability related persuasive systems (Lockton et al., 2014). In this regard, using natural spoken language (i.e., in interviews) may offer a quite promising and unbiased information source to formalize habits (Preece et al., 2015).

To take into account the high individuality of work-related habits and simultaneously include this information in a user-centered design process and system implementation, our work focuses on the use of qualitative data gathered in interviews during early and mid-term stages of user-centered development (Abras et al., 2004). Our method combines this with the potential benefit of semi-formal description concepts for the design process (Dijkman et al., 2008; Mazanek & Hanus, 2011), in our case the Business Process Modeling and Notation modeling language (BPMN) (White & Miers, 2008). Modeling methods in software development enable discussion and documentation of design decisions in the development process (Völter et al., 2013). Modeling may also create the basis for the implementation of prototypes for the iterative development in user-centered design or even can be directly utilized for implementation in case executable descriptions are used, such as Petri nets (Navarre et al., 2009; Weyers et al., 2017, 2020). For instance, Law et al. (2017) present a mapping approach of BPMN models to Petri nets aiming at the model-based implementation of persuasive systems. However, to create meaningful and useful models, a modeler needs expertise and skills, including being able to abstract from given information or knowledge (Weyers et al., 2017), which cannot be assumed from a non-expert modeler.

Therefore in this work, we investigate a method that supports the transformation of qualitative data gathered in an interview-based approach into a semi-formal description concept (BPMN). This method allows the involvement of non-expert modelers and promises to fill the gap between qualitative interview data and models. Eventually, we are aiming at a faster and more stable mapping of interview data to models, which can be used in software development processes for interactive systems. Additionally, we hope to lay the basis for an approach to individualize persuasive systems for formalized end user models and behaviours by operationalizing such gathered models in the actual system implementation, i.e., Langel et al. (2018).

We are presenting our approach by means of how habits, especially work-related habits, can be modeled as processes with the help of semi-formal modeling languages, i.e., BPMN. Based on the work presented by Law et al. (2017), we therefore use CURRENT and TARGET habits, orally described by the office workers themselves to track their activities and trigger alternative actions where needed. In order to achieve this, we asked office workers in a study to imagine situations, in which they have shown the unwanted habit and asked them to use the thinking aloud approach (Ericsson & Simon, 1993) to describe this habit as well as an alternative behaviour they would like to show instead. We will refer to these interviews as thinking aloud interviews (TAI) in the following, since in the interviews we asked the participant to think aloud about their habits and the related context giving few guiding questions. The method we present is then applied to transform the descriptions resulting from TAIs into BPMN models following structured transformation steps. The method represents a structured transformation using a guidebook described in more detail in Section 3 below.

Both representations, TAI and BPMN, pose advantages and shortcomings. On the one hand, TAI lets the end-users freely express their ideas for a more comprehensive and detailed description of the habit, but the unstructured verbal protocols resulting from the TAI process can be overwhelming to a modeler and do not provide a format that can be easily integrated into an interactive system. On the other hand, BPMN diagrams are structured, but might bias the modeler into modeling a habit as rigid sequences of tasks and events, which can result in the omission of important context information that is necessary for understanding the specific habit and for later stages of implementation. Furthermore, the unstructured nature of the input data might result in non-deterministic results due to the style of a certain modeler. In contrast, we expect to take advantage of the positive aspects of TAI and BPMN by using them in a user-centered design process by starting with TAI and subsequently transforming them into BPMN by means of the previously mentioned structured method. Thus in conclusion, the major contribution of this work is two-fold:

- We present a new method for semi-automatic transformation of informal and qualitative data emerging from TAIs into BPMNs (see Section 3).
- We present an empirical evaluation of this approach that shows that our method is suitable in terms of allowing non-expert modelers to learn to create better models (compared to models created by expert modelers) and to reduce variability among models created by different non-expert modelers (see Sections 4–6).

2. Background & related work

This section will give an insight into the background of the presented work in terms of discussing habitual behavior in regards of task modeling approaches and notations as applied in interactive system design and introducing concepts such as think aloud protocols and interviews (Bowen et al., 2021). In addition, related work and state of the art in terms of model creation from qualitative data in the light of process model creation and analysis will be discussed.

2.1. Modeling habitual behavior vs. task modeling in interactive system design

Habits are formed as the result of a process by which repetition of a stimulus-response association generates an automatic reaction (Gardner, 2015). In this regard, the work presented here considers worker tasks and activities at their coarsest level, and how and why workers switch from activity to the next, without going into the details of how each activity is performed. This contrasts other process modeling methods, such as Hierarchical Task Analysis (Annett, 2003), where the approach is to decompose the tasks into smaller operations to understand how the task is performed. Thus, classic task notations such as CTT (Paternò, 2004) or Hamsters (Martinie et al., 2011) seem not to be suitable as we focus on a sequential and imperative description. As a result, we decided to use the Business Process Modeling Notation (BPMN), as it is a widely used process oriented description language, offering a visual representation and a broad variety of tools (White & Miers, 2008).

2.2. Thinking aloud protocol and thinking aloud interviews

Thinking Aloud is a method used to gather verbal data, whereby persons verbalize their thoughts and emotions while working on a particular task (Boren & Ramey, 2000). An important aspect of the method is that participants verbalize whatever comes to their mind. Verbalizations while working on the task are audio-recorded and later transcribed. The transcribed protocols are called verbal or thinking aloud protocols (Ericsson & Simon, 1993; Jääskeläinen, 2010). For instance, thinking aloud protocols can be used for comparing performance and understanding problem solving processes among experts (Sonnentag, 1998), assessing reading comprehension (Magliano & Millis, 2003) or evaluating the usability of certain applications (Boren & Ramey, 2000). The basic assumption underlying the interpretation of thinking aloud protocols is that participants only verbalize information brought into short-term memory by the cognitive processes accompanying the work on the task itself (Ericsson & Simon, 1993). The stream of verbalization reflects cognitive processes and also offers information about the representations participants have (Trickett & Trafton, 2007). In our case, the verbalizations refer to workers' representations of their individual (habitual) behavior and the situation the behavior is embedded in. However, workers are not actually working on the task or showing the habit at the very moment during the interview. Instead, we asked workers to envision a specific situation in which they show the habit and cognitively walk through this situation and verbalize it in detail. This approach is similar to the cognitive inspection of a task to examine a given interactive system (Lewis & Wharton, 1997). With very small additional verbal directives given by the interviewer, the workers can refocus their descriptions on a process-oriented verbalization. Thus, we aim at a verbalization of habitual behavior in an as natural and open way as possible. With this approach, we hope for an unbiased description of habitual behavior—in contrast to pre-defined interview protocols or structures imposed on the descriptions, which may mask relevant information about the actual habitual behavior. We will refer to this method as thinking aloud interview (TAI, see above), because it represents a blend of think aloud and interview methods.

2.3. Model creation from TAI data

After obtaining the protocols, the next step is to extract information and analyze the seemingly qualitative data in a structured way. In this stage, it is important that the person/modeler who analyzes the protocol does not add their own interpretation of what is described in the protocol to the final model (Trickett & Trafton, 2007). One difficulty that is encountered here, is that the model of what is being described is not known a priori (i.e., the modeler does not know the exact habit that is being described); and in fact, the goal of the analysis is to build a model of this process. As a consequence, part of the analysis task should include recognizing what are the relevant pieces of information in the TAI, thus adding a layer of complexity (Chi, 1997).

As a target notation, we used the BPMN. The BPMN standard was developed to help teams made of technical and non-technical members to specify business processes with a common graphical language (White & Miers, 2008). It consists of a series of elements that represent user and system tasks, events, and flow structures such as conditional and parallel bifurcations (gateways). Moreover, it has being demonstrated that BPMN diagrams can be transformed into Petri nets (Law et al., 2017; Petri, 1966), which can be easily integrated into interactive systems (Weyers et al., 2011, 2012, 2017). The method presented here parts from the different work on verbal protocol analysis and applies it to specific accounts of unstructured and informal work-related habits and takes them a step further by offering a transformation into BPMN that can be even further transformed into more formal structures. More light on the general challenge of handling qualitative data is shed in the next section below, which also applies to the method presented here.

2.4. Handling qualitative data for model creation

To develop our method to transform textual qualitative data into BPMN models, we based our approach on existing work on qualitative data analysis. Qualitative data is used to gain a deeper understanding of processes, systems, behaviours and problems in areas such as healthcare (Bradley et al., 2007; Smith & Firth, 2011), business (Mayer, 2015) and risk management (Kelly (Letcher) et al., 2013). However, processing and analyzing qualitative data can be a complex task and has to be done systematically and carefully to avoid pitfalls. These pitfalls may include issues such as overgeneralizing results (Neusar, 2014) or creating false narratives due to cognitive bias and overlooking important information (Gilbert et al., 2014) due to the quantity and complexity of data.

Various authors present processes, tasks and steps for qualitative data analysis (Liamputtong, 2009; Srivastava & Hopwood, 2009; Thomas, 2006; Welsh, 2002), with Mayer (2015) providing an exhaustive review of methods and approaches. For verbal protocols (such as TAIs), some work on analysis and coding such protocols exist. Lemke (2012) applies the analysis on different dimensions: structural, semantic and rhetorical. Chi (1997) presents a process that consists of eight steps. Roughly, the idea is to search for concepts and relations in the verbal protocol and create conceptual maps from which patterns can be discovered. The method we present here is based on these steps, but simplified and adapted to obtain a more concrete process specific for transforming TAIs into BPMN models. The specific application of the method depends on factors such as data type and quantity, complexity and goals of the analysis. However, similar steps can be identified throughout the different methodologies that are available and condensed into the following:

1. *Explore the data*: This step consists of reading, understanding and organizing the data. It may also include displaying the data in graphs or diagrams to facilitate an overview of the data and to organize it.
2. *Reduce*: This is the step where the actual analysis takes place. In this step the modeler tries to find patterns, codes and themes.
3. *Conclude*: In this step conclusions from the analysis are drawn, connections are created, hypotheses are formulated or tested (depending on the stage of the research).

The process is always iterative and cyclic, after finishing the last step, the process begins again by revisiting the data in the light of new conclusions and observations. The general agreement is that the process, in any of its different flavors, is a lengthy one, requiring also expert knowledge of the methodology employed. Data, codes, concepts and theories emerge after a number of iterations and they are revised and refined with every new iteration.

These works mostly present general guidelines to handle and process qualitative data but do not consider more formalized procedures, leaving many details open to expert criteria. Mauthner and Doucet (2003) describe that qualitative data analysis cannot be separated from the researcher's personal experience, and thus, the process requires expertise. We argue, however, that a structured approach to the analysis may also have the chance to enhance reliability of results. Additionally, expert modelers are not always available to process and analyze the amount of data that is being generated and training also demands a great effort. Hence, for the purpose of our research, we condensed the iterative process of analysing the data and created a method to process new data, that can be applied by non-expert modelers (with experience in modelling) to facilitate and shorten the process of extracting relevant information from textual data obtained in interviews. With this, our approach aims at faster turn-arounds in the user-centered design process as well as the potential adaption of a persuasive system to the end-user's specific habit.

2.5. Process model creation and evaluation

In the context of BPMN process model creation and analysis, a large body of work has been established over the last decades. Algorithmic as well as manual approaches have been presented to gather process models from various types of input data. In terms of algorithmic approaches, for instance, works by van der Aa et al. (2019), Rebmann and Van der Aa (2021), or van der Aa et al. (2020) present methods to gather declarative models from textual and natural language information. These works generate declarative models (in most cases constraint-based descriptions) rather than imperative descriptions (such as BPMN) based on either natural language to generate constraints (van der Aa et al., 2020, 2019) or event logs (Rebmann & Van der Aa, 2021). However, in our work we focus on imperative models (BPMN), due to the reasons discussed above.

Work that *generates imperative models*, such as BPMN, have been proposed by Schäfer et al. (2021), which focuses on automatic processing hand-written sketches of BPMN models, or work by Friedrich et al. (2011) and Schumacher et al. (2013), which use natural language in form of text (e.g., from text books or manuals) to generate such models. In both approaches, the assumptions on the input data is quite strong compared to our work, which uses quasi unstructured interviews (TAIs) as input that is transcribed into text for translation. This textual information includes a high level of noise and inconsistencies, which may be only addressable with complex algorithms. Thus, we decided to develop a structured method reducing the complexity of transformation, reducing the influence of interpretation of the modeler transforming the input data but at the same time keeping a human in the loop to take advantage of their ability to understand complex natural language.

In these terms, another stream of research can be identified on the *investigation of the actual model creation process*, such as presented by Pinggera et al. (2012). These authors present work on the analysis of model creation processes with a high relevance especially in terms of BPMN and process models in software development. They identify different styles in the creation of business process models ("(1) modeling with high efficiency, (2) modeling emphasizing a good layout of the model, being created less efficiently, and (3) modeling that is neither very efficient nor very focused on layouting" (Pinggera et al., 2012, p. 1056)). However, in our work we focus on the investigation of a pre-defined creation process and, thus, focus more on the quality of the actual created model by applying this method (as discussed above). Similar work has been presented by other authors (Claes et al., 2012; Pinggera et al., 2012; Weber et al., 2015, 2016) who focus on cognitive measures to investigate the actual modeling process, which is not the focus in this work but may be addressed in future investigations on the actual modeling process conducted by the modelers applying our method.

In terms of *model quality*, various works have been presented (Bolle & Claes, 2019; Gruhn & Laue, 2007; Krogstie, 2012; Vanderfeesten et al., 2007). Most of these works consider qualitative requirements due to the created models, such as "Customer Value" and "Economic Profit" or "Employee Well-Being," which have been measured by, for instance, accuracy, specificity to target, or balancing cost vs. quality and speed vs. flexibility (Krogstie, 2012). However in our work, we actually focus on similarity of created models using our method (see Section 4) and thus use similarity measures but neglect more abstract measures due to qualitative assessments such as in Krogstie (2012). For future work, this perspective on created models will be of high relevance, specifically when created models are more commonly used in persuasive systems and its effectiveness needs to be measured and considered also due to the actual model of the habit used.

3. Method

The main goal of the proposed transformation method is to obtain BPMN models from TAIs with descriptions of work-related habitual behavior, in a way that interpretation-bias from the modeler is minimized. This should lead to more consistency throughout modeled diagrams, which is empirically investigated in Section 4 below. Additionally, by following our proposed steps, expert knowledge in BPMN modeling should not be required to obtain correct and consistent results. The transformation steps were based on TAI analysis procedures such as presented by Chi (1997). Following this, our first two steps were to separate the text in smaller units (Segmentation). Each segment is then labeled with one of six classes, which are roughly based on BPMN elements (Classification). The classified segments are finally ordered and grouped into BPMN elements and connected to each other creating a final BPMN model (Modeling). The created manual including the following description and material is available as part of the raw data publication accessible via doi:10.5281/zenodo.4704593 on the Zenodo platform.

Thus, the proposed transformation method consists of three steps:

1. Segmentation: Separate the TAI into smaller units, using verbs as guides.
2. Classification: Categorize each of the segments into one of a set of classes.
3. Modeling: Convert the classified segments into a BPMN model.

These steps consist of and are executed by the modeler following a number of rules that are described in detail in the following subsections.

3.1. Segmentation

As mentioned before, in this step, the original transcription of the TAI interview is divided into segments. Here, verbs are used as an indicator of each segment (Figure 1, rule 1), as they generally relate to tasks and most events. Exceptions to this rule are relative and adjective clauses (Figure 1, rule 2), which should not be separated, and time related segments (Figure 1, rule 3), which should be separated even if no verb is present. In all other cases, and whenever there is doubt, a finer segmentation is recommended (Figure 1, rule 4).

In general, segmentation is a difficult task because there are no established boundaries that define the units, which leads to ambiguities that are then subject to interpretation of the modeler. In an optimal scenario, each segment should map to one Task, Event or relation between these, so that a BPMN graph can be easily modeled. However, due to the freedom in which TAIs are gathered, many segments can make up, for example one Task, since the granularity is not predefined. For instance, if a TAI describes a telephone conversation in detail, e.g., "dialing," "saying hello," "asking questions," "answering questions," "saying goodbye," a modeler *A* may decide to create a task with the label "telephone conversation," whereas a modeler *B* breaks down the task and creates an element for each part. Thus, the level of detail of descriptions varies between modelers. If segments are too large (comprise many details), we have the risk of losing or mixing information, which makes the next step "classification" harder to complete. Therefore, we opted for rather smaller units indicated by verbs rather than punctuation (periods or commas), that can be then regrouped in retrospective, once the classification step has been completed. Figure 2, part A, shows an example TAI. Part B shows the corresponding segmented text, ordered into a table. Verbs are shown in bold. Notice that the fifth segment contains 2 verbs, since the second is part of a dependent clause, which should not be segmented.

3.2. Classification

In this step, the task is to label (the class) and number each segment according to a given classification. Rather than describing the details of how a specific task is performed, our models should describe the situations (relations between activities and events) that lead to triggering habitual behavior at the work place. Thus, the proposed method does not have semantics for artifacts such as tools, data or other actors. Instead, the main artifacts are events, tasks and conditions. For the classification, we identify six classes: Setting, Annotation, Task, Event, Condition, and Other. These

1. Segments are defined by the presence of a verb.

2. Exceptions are relative and adjective clauses that give more information on a subject.

3. Parts that describe time or time periods are segments too.

Applying this rule may result in segments without a verb.

4. A finer segmentation is preferred.

Figure 1. Segmentation rules as presented in the transformation manual.

A. TAI

I was working on my travel expenses report and I received an Email. It was a reminder to register to a conference. I needed to do this immediately. After this, I didn't know anymore where I was with my my travel expenses report, so I had to start over with it.

B. Segmentation and Classification

I was **working** on my travel expenses report	T1
and I **received** an Email.	E1+
It **was** a reminder to register to a conference.	AE1
I **needed to do** this immediately.	T2
After this, I **didn't know**anymore where I **was** with my travel expenses report,	S
so I **had to start** over with it.	T3

C. Model

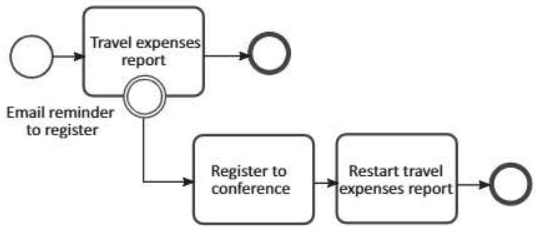

Figure 2. TAI to BPMN transformation example. (A) TAI raw text. (B) Segmentation and classification step. (C) Resulting BPMN Model.

classes were based on the different elements we found in the descriptions and selected so that they roughly match the elements used in BPMN modeling. *Setting* refers to sentences where the interviewed workers described their surroundings, workplace or mood (Figure 3, rule 5). *Task* refers to any action performed by the interviewed worker (Figure 3, rule 7). *Parallel tasks* have to be marked additionally with a plus (+) sign (Figure 3, rule 8). On the other hand, Events are those incidents that happen in the environment (Figure 3, rule 9) or actions from other actors (Figure 3, rule 10). Events include also spontaneous reactions and thoughts (Figure 3, rule 11) and descriptions of time moments or periods (Figure 3, rule 12). We also make a distinction between events that happen during the execution of a task and events that happens in-between tasks (Figure 3, rule 13). A *Condition* is any content that indicates that an alternative path of action might be taken (Figure 3, rule 14). Segments classified as *Other* do not fall under these classes, such as rhetorical expressions (Figure 3, rule 16) or correspond to segments produced by the interviewer (Figure 3, rule 17). Figure 2, part B, shows the classification for the segments of our example (right column).

Note that during classification, segments might be regrouped by using the *Annotation* class (Figure 3, rule 10). For example, when the description of a task spans multiple segments, each additional segment will receive an annotation label with a reference to the main segment. Thus, the regrouping happens implicitly during classification.

An issue to consider here is the modeler's interpretation. Some words or structures used by the interviewed workers, for instance when technical specialists use a specific jargon, can mean something different to the modeler. The modelers should then restrain from bringing their own views into the transformations but only use the literal information that is present in the text. For example, considering the following text "I don't like to answer the phone if I do not recognize the number," someone could wrongly deduce that the person does not answer the phone when the caller is not recognized (since they do not like to do it). However, this information is not really present in the text, it can only be assumed by the modeler from personal experience and preference. From only this text, it is not possible to know for certain if the person does not answer at all or just does it unwillingly. The correct interpretation, if no other information is available, should reflect this lack of information and classify the segment accordingly, in this case, as Setting or Annotation.

3.3. Modeling

For the modeling step, segments classified as Tasks, Events or Conditions are directly converted into the corresponding BPMN elements and subgraphs (Figure 4, rules 1–3). Events that happen during the execution of a task are converted to boundary events (as defined in the BPMN standard, also see Figure 4, rule 4a). Tasks that were identified as parallel, are enclosed by parallel gateways while Conditions are branched with exclusive gateways (Figure 4, rules 5 and 6). After this conversion, if it turns out only one of the paths in the exclusive gateway was described (Figure 4, rule 5a), the gateway is substituted with an Event. This is done to avoid having incomplete diagrams. All other segment classes are used only as reference and information and do not directly translate into a BPMN element or subgraph. These can, however, be added as annotations in the BPMN model. These rules are further extended by the notes shown in Figure 5, which clarify some general aspects of the modeling step. Figure 2, part C, shows the final transformed BPMN Model of our previous example. The three tasks and the event can be traced one-to-one into this model; the Setting segments and Annotations do not appear in the final model but maybe used for additional annotation.

4. Evaluation

To validate our transformation method and to examine if our proposed method had an effect on the quality of the BPMN models derived from the TAIs, we conducted an experimental study with an experimental and a control group (independent group design). For validation, we aimed at showing that (i) applying our method reduces variance in the resulting models and thereby reducing individual interpretation of modelers, as well as (ii) persons with experience in modeling but without specific knowledge in TAI and BPMN (non-expert modelers) are able to execute the transformation with higher quality than without the manual. The two variables measured are then (i) the difference between the variances among the solutions within the experimental and control group and (ii) the difference of the average quality of the solutions within the experimental and control group with respect to expert solutions.

Therefore, the participants in the experimental group were instructed to use our transformation method, participants in the control group performed the transformations freely (without any specific method).

(a) ***Setting (S):***

5. General descriptions of the situation: When and where it happened, and feeling and emotions of the person when it happened.

(b) ***Annotations (ARef)***

6. Similar to Setting, but applied specifically to a Task, Event or Condition.

Task (Tn(+))

7. Any activity performed by the user.

8. (+) Parallel Tasks: Some tasks may contain words or phrases that suggest the task is performed at the same time as another one.

Event (En(+)):

9. All incidents in the environment.

10. All actions through other actors.

11. Thoughts, feelings and emotions from the user that happen spontaneously.

12. Time or time periods.

13. (+) Events that occur during the execution of a task

(c) ***Condition (C):***

14. Description of alternatives, for example yes/no questions that are often introduced by "if"

15. Time or time periods (time-dependent conditions).

Other (On):

16. All segments that are not directly part of the description.

17. All segments that were produced by the interviewer.

Figure 3. Classification rules as presented in the transformation manual.

According to our major study goal, we assumed that among participants in the experimental group, the BPMN solutions developed by using our transformation method would be more similar to each other than among participants of the control group. In other words, we expected that participants within the experimental group would converge towards similar solutions, whereas participants in the control group would not. If the data supported this assumption, then we can conclude that the transformation method helps to reduce the variance in the interpretation of the TAIs and, accordingly, the variance in the resulting transformations and solutions. As a

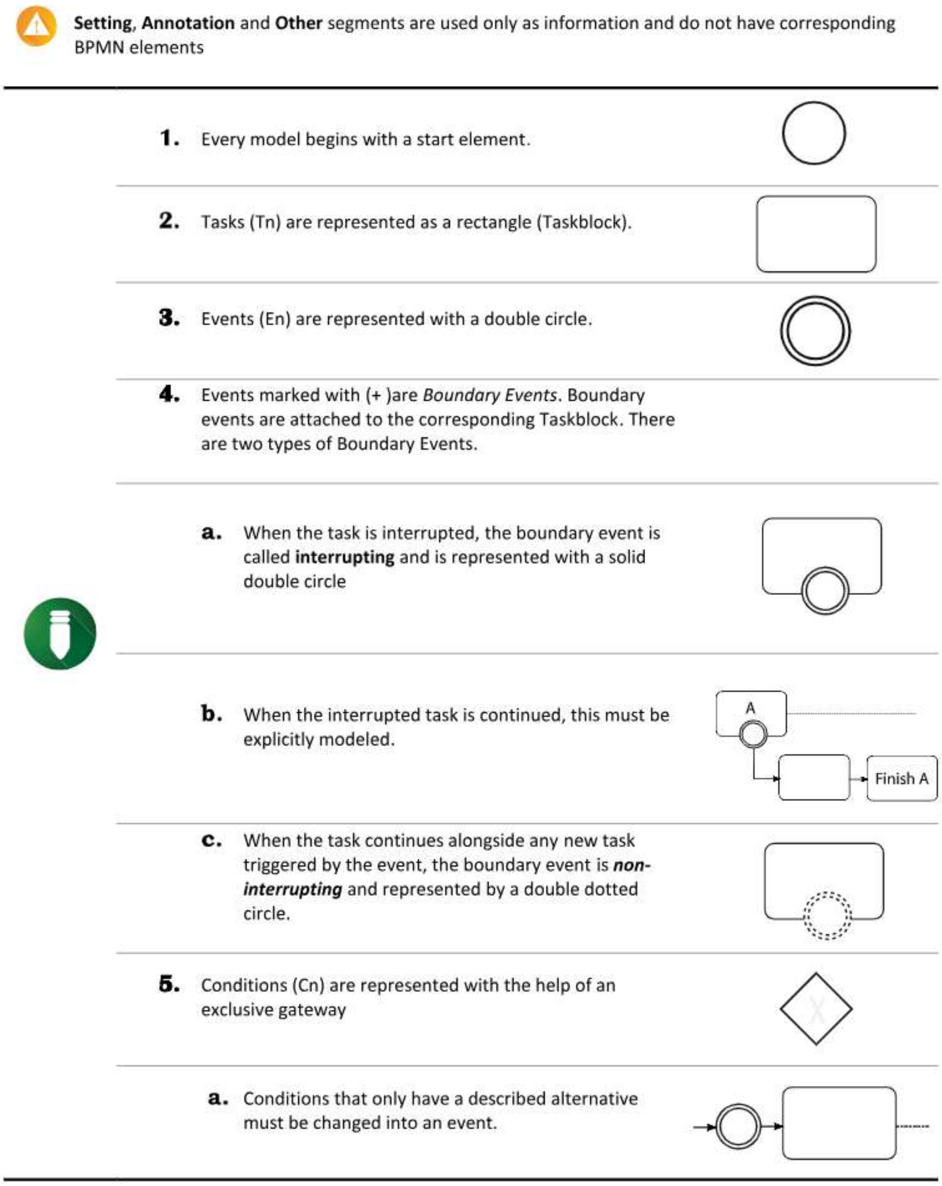

Figure 4. Modeling rules as presented in the transformation manual (Part 1).

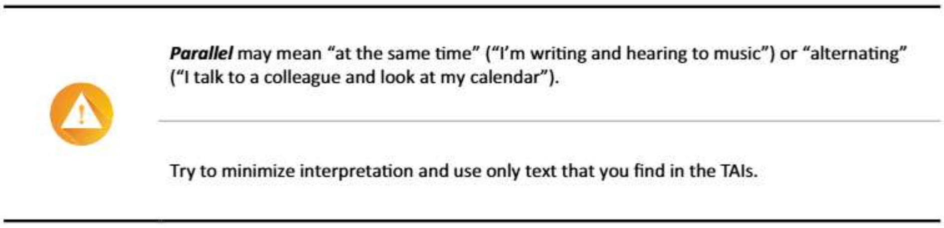

Figure 5. Modeling notes as presented in the transformation manual.

precondition, we hypothesize that the solutions of both groups will be different.

Hypothesis 1: The solutions generated by participants in the experimental group will be different from the solutions generated by the participants in the control group, with the solutions within the experimental group being more uniform (i.e., more similar to each other) compared to the solutions within the control group.

To further elaborate if the use of the manual also contributed to increased quality, we expected that the BPMN solutions developed by participants in the experimental group should also be more accurate and correspond more closely to an expert solution.

Hypothesis 2: The solutions of the experimental group will be more similar to expert solutions than the solutions of the control group.

4.1. TAI data acquisition

In order to obtain data to work with, as examples and for later evaluation, TAIs were collected by conducting interviews with employees working in jobs involving administrative and organizational tasks. In total, 30 employees were interviewed and each described one habit in two forms: in its CURRENT version and how they wished to change it, the TARGET, resulting in 60 interview protocols. The interviews took between 8 and 24 min and were audio-recorded. The specific process was as follows:

1. Participants were asked to report one to three work habits, and to evaluate them in terms of certain criteria, such as frequency or changeability (Verplanken & Orbell, 2003).
2. Based on the criteria, one particular habit was chosen for the TAI (the CURRENT habit).
3. To help participants remember the last situation where the work habit occurred, they were asked to describe situational or psychological aspects of the situation (e.g., "Was it a calm environment or rather loud?," "How did you feel?").
4. Participants were asked to describe the situation, underscoring that they should not self-censor their utterances in any form, describe as concrete as possible from a first-person perspective and try to consider the chronological order of events in their description. Also, participants were asked to think aloud and describe their behavior, thoughts and emotions (in the situation) as precisely as possible.
5. If the interviewer had the impression that the description did not allow for a sufficient understanding of the chronological order of events, important aspects of the situational environment, or the habitual behavior itself, he/she would ask for these aspects using open questions (e.g., "Could you explain the chronological order of events one more time?," "Could you describe the environment in more detail?," "Could you explain what in this situation represents your habit in more detail?").
6. Participants were encouraged to describe the situation again, but this time by imagining themselves behaving in a way handling the habit and thus rendering it unproblematic or at least less burdensome (the TARGET habit).

Participants were also asked to model the same habit using BPMN in a participative manner, with the help of the interviewer. To counter the learning effect, the order of the collection method (TAI or BPMN) was randomized.

All 60 audio-recordings were then transcribed by a person different than the interviewer. Two of the authors of this work then classified the TAIs according to their complexity. For this, they identified three criteria: the volume of text, if the way the participants expressed themselves was clear or not, and if the description of the situation and the events were presented in a chronological order. With these criteria, the TAIs were classified into 3 levels of complexity. From the 60 TAIs, 25 had a low-complexity level, 28 had a medium complexity and 7 had a high complexity.

Two of these TAIs (referred to as TAI 1 and TAI 2 in the following) with medium complexity were selected to use in the evaluation presented here. To reduce the duration of the study, these TAIs were shortened, by cutting text at the beginning and at the end only. This text was mainly contextual explanations (from Step 3) and did not include any description of the habit. No paraphrasing or similar editing was done. The resulting TAIs had 50 (TAI 1) and 25 (TAI 2) segments (according to the segmentation step of the proposed method) respectively.

4.2. Pre-study

To test the feasibility and duration of our study procedure and to examine potential problems within the transformation manual, we performed a pre-study with ten computer science students as participants who did not have any prior knowledge of the transformation method. Participants in the pre-study were provided with six different TAIs to work on. Each participant only had to apply one of the three transformation steps to any given TAI, i.e., either segmentation, classification or BPMN modeling. Thus, each participant had to solve six tasks, two for each one of the transformation steps. In Task 1 and Task 2, participants were asked to segment a TAI by placing marks to separate the text. For Task 3 and Task 4, participants were provided with a segmented TAI and were asked to assign a class to each of the segments. Finally, for Task 5 and Task 6, the participants had to construct BPMN diagrams based on already segmented and classified TAIs. The results of the pre-study showed that most differences between participants occured in the classification and modeling steps, while the segmentation step appeared to be straight-forward, with most of the participants producing similar segmented texts. Additionally, we detected some minor errors and confusing statements in the study material that we corrected for the main study.

4.3. Participants

Forty-one (6 female, 34 male, 1 no answer) undergraduate students participated voluntarily in the study. All participants were recruited via a recruiting platform available at the university and had no direct relation (e.g., student teacher) with the authors. The mean age of the participants was 22.75 years ($SD = 3.4$). All participants majored in a computer science or a related study program. As an incentive for participation, a small monetary compensation was given to each participant. Six participants reported having at least heard of BPMN, 5 had learned about it in a lecture, and the remaining 30 had no previous knowledge. Thirty-seven participants reported knowledge on UML or Flowcharts, ranging from "have heard of it" to "used it in student projects"; the remaining 4 reported no experience in any of the 3 or other modeling languages.

4.4. Materials

Participants in *both* experimental and control groups received:

- An explanation and introduction into BPMN modeling with examples; this material was given in printed and digital versions.
- Two pre-segmented TAIs. Based on insights gained during the previously described pre-study and to shorten the time required to finish, we decided to test only the classification and modeling steps. To avoid bias in the input, the same segmented TAIs were given to both groups.
- Printed instructions for completing the experimental tasks. The corresponding paper with a marked space for reporting the solutions was provided as well.
- For completing the modeling tasks, participants could use a whiteboard and magnets with the BPMN elements that they could move around easily. However, the final solutions had to be provided on paper. The use of the magnets was optional and not part of the evaluation.
- A post-study questionnaire which assessed participants' perceived confidence in their results and performance. With this subjective assessment, we aimed at supporting the objective measures with the participants' perceived effectiveness of the method.

Participants in the *experimental group only* received (additionally):

- A transformation manual in digital form, with the rules as presented in Figures 1,3,4,5,6, including examples and exercises. The exercises presented after the introduction of each step had to be solved before participants could proceed to learn about the next step of the method. Participants could check the correct answers and could access an explanation immediately after providing the answer.
- A printed version of the manual (without the exercises) to use as reference during the task if desired.
- A post-study questionnaire to evaluate the quality of the provided manual and exercises and the helpfulness of the proposed method.

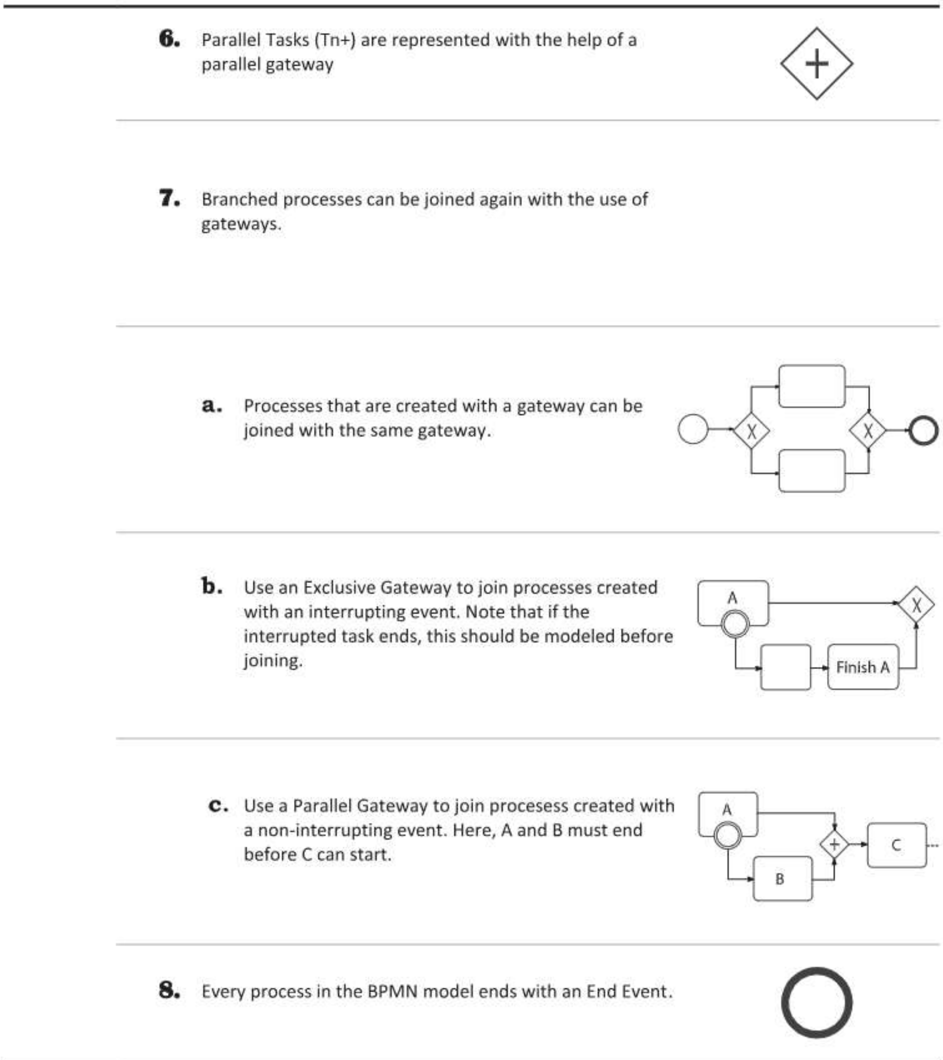

Figure 6. Modeling rules as presented in the transformation manual (Part 2).

The material is available under doi:10.5281/zenodo.4704593 on the Zenodo platform.

4.5. Procedure

The main study took place in a room where the participants could sit comfortably and work individually. Participants were randomly assigned to the experimental or control group at arrival. To start, participants signed an informed consent, followed by a brief introduction to the study and a demographic questionnaire. This part took around 10 min. For the experimental group, the study continued with the tutorial that included all the steps of the method, corresponding exercises, and the introduction into BPMN. For the control group, this part consisted only of the introduction into BPMN. When the participants had finished, they continued with the study tasks. The completion time for these parts together varied between 60 and 90 min. After completing the tasks, the participants had to answer the questionnaire regarding the method (for the experimental group) and the material provided, which took around 15 min. In total, the complete procedure took between 1.5 and 2 hours to complete. The procedure was carried out entirely by one of the authors for all 41 participants.

5. Results

In the following subsections, we report the study results. The evaluation reports are divided into four subsections. First, we report the analysis method applied to the modeling results (5.1), which enabled us to analyze quantified data in a second step. These quantified data is then presented in the light of our two hypotheses. Finally, we will review some of the most relevant results of the post-study questionnaire presented in Sec. 5.4. The raw data is available under doi:10.5281/zenodo.4704593 on the Zenodo platform. The data includes the study's raw data, documentation for the analysis as well as study material (including informed consent, task descriptions and material, questionnaires, and the website used for training) used for the study.

5.1. Modeling result analysis

To analyze the data, we first divided each of the two TAIs (and consequently also each BPMN model developed by the study participants) into smaller, meaningful parts based on the sequential habitual work process reported by the employees who provided the TAIs (see Section 4.1). Therefore, we had several sessions discussing this division in very detail after creating them separately first. For TAI 1, we identified eight meaningful parts (Part A to Part H), and for TAI 2, we identified four meaningful parts (Part A to Part D). For example, in TAI 1, it can be clearly distinguished how the person (i) starts a task, (ii) then makes a pause in which they do something else, (iii) then continue with the task, and so on. Note that we have included parts of the TAI that only describe the setting as well because some participants (in both groups) included settings in their final solutions—although they should not be part of the final solutions in a model created by an expert. We then examined the BPMN models developed by the participants and recorded the solution category (i.e., Task, Event, Condition, Setting, Annotation, Other) that were used by the participant for each of the eight (in case of TAI 1) or four (in case of TAI 2) parts in the participant's model. Then, we tabulated how each participant modeled the eight parts for TAI 1 (labeled from A to H) and the four parts for TAI 2 (labeled from A to D).

5.2. Test of Hypothesis 1

We recorded each different partial solution that participants provided and counted how often participants in a specific group used these solutions. In other words, we examined if in the different groups different concentrations on particular solutions occurred. These concentrations on specific solutions or distributions of solutions are shown in Figures 7 and 8 for TAI 1 and TAI 2, respectively. We ran Fisher exact Chi Square tests to determine if there were different distributions of solutions between the groups. A partially post-hoc power analysis (Dziak et al., 2020) done with G*Power 3 (Faul et al., 2007) for the different parts of the TAI indicated that statistical power for detecting a medium effect ($w = 0.3$) was between .17 and .33 for TAI 1 and between .14 and .30 for TAI 2. Since the statistical power was low, the probability to miss effects is increased. The Fisher's exact Chi Square test shows that there were significant differences between the experimental and control group with respect to the solutions provided for several parts (for TAI1: part D ($\chi^2(7) = 16.229$, $p = .005$), part G ($\chi^2(14) = 21.714$, $p = .007$); for TAI 2: part B ($\chi^2(7) = 14.232$, $p = .008$)). By visually inspecting the graphs in Figures 7 and 8—which are graphical representations of frequency tables—it can be seen that in general for the experimental group (for the parts marked with a star), there is a higher concentration of participants choosing a reduced number of solutions, compared to the control group.

Overall, findings tend to be in line with Hypothesis 1 because parts of the resulting models generated by the experimental group were more similar to each other, compared to the similarity between the solutions generated within the control group.

5.3. Test of Hypothesis 2

To test Hypothesis 2, a ground truth solution provided by two experts familiar with the method was compared with the participants' solutions. One of these experts is an author of the present work who did not conduct the study as experimenter. Thus, experts were blind to the participants' group (experimental group versus control group), were not involved in the data collection and had not seen the participants' solutions beforehand. To create the *ground truth* solution, the experts first created individual solutions, discussed them, and finally agreed upon an ideal solution. Further, relevant parts of the ideal solution were identified. Experts aimed at parsimony, thus their solutions contained as few elements as possible while capturing the relevant parts of

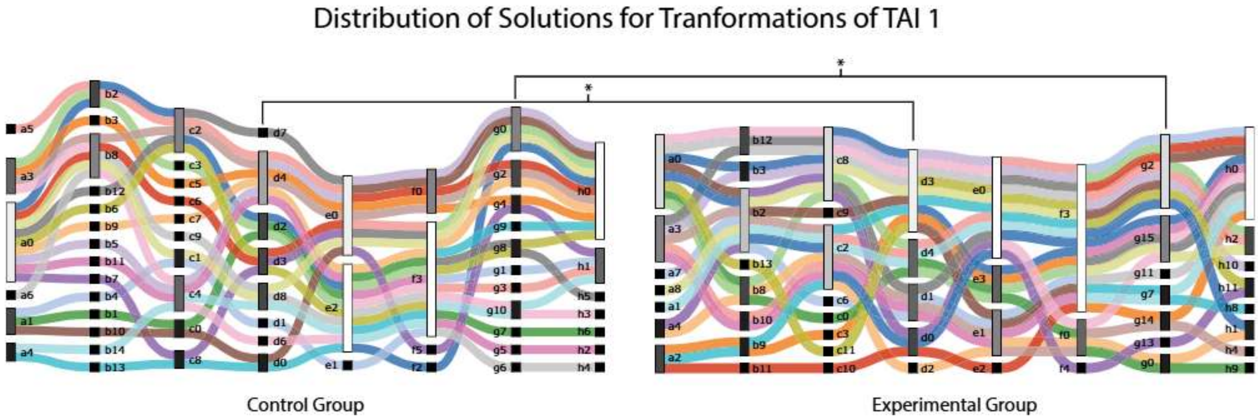

Figure 7. Distribution of solutions for Transformation of TAI 1. Each colored line corresponds to one participant's solution. Each labeled bar corresponds to one solution; the size and color of the bar corresponds to the number of participants that used that solution. Bars labeled with 0 indicate that the part was not modeled. Stars indicate significant differences for a specific part between groups.

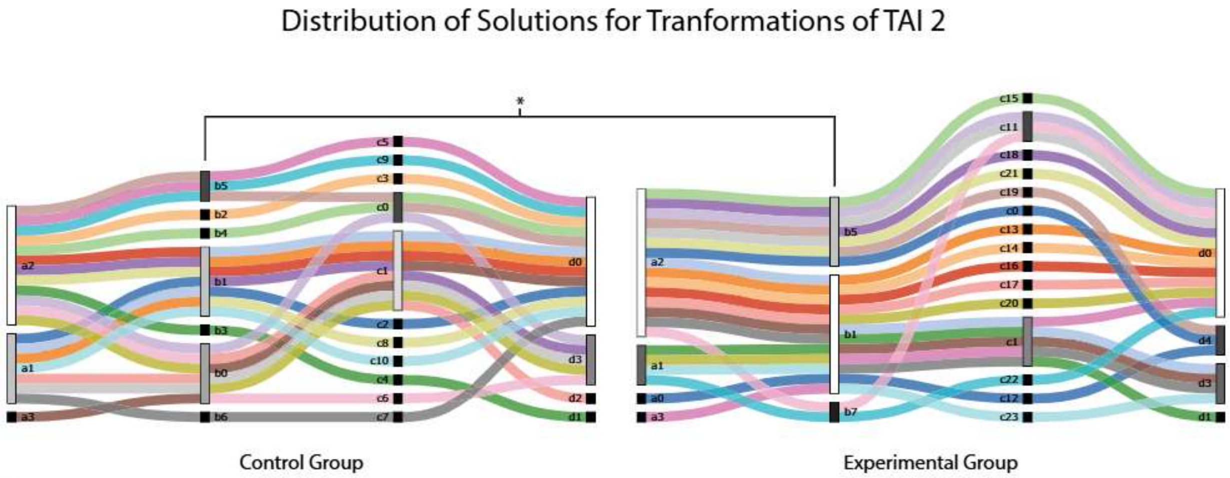

Figure 8. Distribution of solutions for Transformation of TAI 2. Each colored line corresponds to one participant's solution. Each labeled bar corresponds to one solution; the size and color of the bar corresponds to the number of participants that used that solution. Bars labeled with 0 indicate that the part was not modeled. Stars indicate significant differences for a specific part between groups.

the TAIs. Experts solutions had 6 relevant parts for TAI 1 and 3 relevant parts for TAI 2. By comparing experts' and participants' solutions, similarities and differences were identified. Accordingly, relevant parts of the participants' solutions were matched to the relevant parts of the expert solutions (see Table 1).

The experts rated the syntactic and semantic quality of participants' solution. First, syntactic quality was evaluated via expert judgments of how relevant parts (from the ground truth solutions) of TAI's were modeled by participants. Each relevant part could be rated with -2 completely incorrect, -1 mostly incorrect, $0=$indecisive, $+1$ mostly correct, $+2$ completely correct. When the relevant part was missing, -3 was assigned. Second, semantic quality was evaluated via experts' judgements of the whole BPMN solution. In particular, experts rated the semantic quality on:

1. *Level of Detail*: Does the model have the correct level of detail, not more and not less information, compared to the ground truth solution? (-2 = too little information, $+2$ = too much information).
2. *Comprehension*: How easy to understand is the participant's BPMN model? (-2 = very hard, $+2$ = very easy).
3. *Cues*: How many of the relevant cues are correctly represented? (-2 = no relevant cues are correctly represented, $+2$ = all cues are correctly represented)
4. *Interpretation*: How well does the participant's model represent the TAI? (1 = the BPMN does not reflect the TAI at all, 5 = the BPMN represents the TAI correctly)

For analysis, we averaged the scores provided by the two expert raters for syntactic and semantic quality respectively. We ran two-tailed *t*-tests to examine if the experimental group provided solutions with higher quality than the control group. A partially post-hoc power analysis done with G*Power 3 (Faul et al., 2007) indicated that statistical power for detecting a medium effect ($d = 0.5$) was at .35. Given that our procedure represents a first approach to formalize highly informal TAIs, we deem the statistical power satisfactory.

The syntactic quality was significantly higher in the experimental group for part Be ($t = 2.077$; $df = 39$; $p = .044$; $M^{diff} = 0.49$, BCa 95% $CI[.013, .096]$; $d = .64$) and Ge2 ($t = 2.099$; $df = 39$; $p = .042$; $M^{diff} = 1.16$, BCa 95%

Table 1. Part correspondence between parts from Participants' and Experts' solution.

TAI 1	Participants	A	B	C	D	E	F	G	H
	Experts	–	Be	Ce1,Ce2	De	–	–	Ge1,Ge2	–
TAI 2	Participants	A	B	C	D				
	Expert	Ae	Be	Be	De				

The parts label with '-' indicate that this was not in the expert solution. Parts C and G of TAI1 was separated into two parts by the experts Ce1, Ce2 and Ge1, Ge2, respectively. Parts B and C of TAI2 were combined into Be by experts.

$CI[.042, 2.276]$; $d = .65$) of TAI1. Also, the total syntactic quality (mean score of all six parts) was higher in the experimental group than in the control group ($t = 2.178$; $df = 39$; $p = .036$; $M^{diff} = 0.55$, BCa 95% $CI[.039, 1.06]$; $d = .68$). For TAI2, the experimental group showed higher syntactic quality in part Be ($t = 3.047$; $df = 39$; $p = .004$; $M^{diff} = 1.02$, BCa 95% $CI[.034, 1.69]$; $d = .95$). For all parts except the part De of TAI2, descriptive scores for syntactic quality were higher in the experimental group. In two additional two-factor ANOVAs on overall syntactic quality of the two TAIs, we did not find any significant main effects of the raters, nor a significant interaction effect between rater and group.

The semantic quality for Comprehension was marginally significantly higher ($t = 1.807$; $df = 39$; $p = .078$; $M^{diff} = 0.63$, BCa 95% $CI[-.075, 1.33]$; $d = .57$) and significantly higher for Interpretation ($t = 2.360$; $df = 39$; $p = .023$; $M^{diff} = 0.76$, BCa 95% $CI[.011, 1.41]$; $d = .74$) in the experimental group for TAI1. For TAI2, there were no significant differences between the groups. Again, in two additional two-factor ANOVAs on overall semantic quality of the two TAIs, we did not find any significant main effects of the raters, nor of the interaction between rater and group. The graph showing the average scores for both TAPs can be seen in Figure 11.

Overall the models provided by the experimental group tended to be of higher syntactic and semantic quality, in particular for TAI1, which was a longer TAI and resulted in a model with more parts and elements. Thus, according to expert criteria, study results tend to be in line with Hypothesis 2.

5.4. Post-study questionnaire

The post-study questionnaire contained 38 questions that covered, among other aspects, participants' confidence and self-assessment of their performance and method evaluation (quality and usefulness; only for the experimental group). Thirty-four of the questions were in the form of statements with a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). The remaining 4 questions were open questions. Here we present a sample of interesting results, shown in Figures 9 and 10. A Mann-Whitney test between both groups shows only a significant difference in the distributions of question Q8: *Previous basic knowledge in modeling is required for the study tasks* ($p = 0.03$). The median value in the experimental group was 6 (mostly agree) and 4 (neutral) in the control group. With respect to the manual itself, a One-Sample Wilcoxon Signed Rank Test within the groups, shows that (compared to the expected median (4.0)) the participants in the experimental group agreed that it was helpful ($M = 5.0; p = .001$) and that the language used was not too technical ($M = 2.0; p = .007$) (questions Q30 and Q31). In general, the example TAIs were found to be very easy to understand compared to the actual TAIs ($M = 6.0; p = .001$) (Q36), but were nonetheless helpful ($M = 5.0; p = .001$) (Q32). Most participants agreed that the classification was difficult ($M = 6.0; p < .001$) (Q12), although only considered class "Condition" as not easy to recognize ($M = 3.0; p = .005$) (Q20). Finally, participants reported relatively low confidence about the answers they provided for the classification ($M = 3.0; p = .026$ $ZM = 3.0; p = .026$) (Q1). Participants in the control group were more confident in their models ($M = 5.0; p = .003$) (Q2).

6. Discussion

Segmentation, as well as classification, are critical steps in the sense that the mistakes done here will carry over to the later steps. Based on the obtained results in the pre-study, we concluded that the classification step was unsurprisingly the most difficult step. This is again confirmed by the provided solutions of the study and by the post-study questionnaire in which most of the participants agreed that this step was difficult. It is also possible to observe that participants did not recognize the annotations easily, although these are normally directly after the main segment. An explanation to this is that some participants felt the segment was too general and preferred to mark it as Setting instead. Another possible explanation is that the granularity of the Task or Event is at this point still not clear, and as a consequence, participants did not actively think about the possibility of using this particular classification to group similar segments together, and instead preferred to use new labels.

The final results show that the models developed by the experimental group have some differences to those developed by the control group. We are confident that with more training on the method, the observed differences would be even larger. Regarding the correctness, we found that some parts were significantly more correct in the solutions provided by the experimental group, compared to the expert solution, supporting *Hypothesis 2*. Looking at Table 1 and Figures 7 and 8, we can also see, that not only were some parts of the models significantly different for the experimental group, but when they were different, the parts were also better rated by the experts. Furthermore, according to the complexity classification performed during the data preparation process (Section 4.1), TAI 2 was more complex than TAI 1. This fact had an impact on the quality of the transformations for this TAI, too, as reflected by the results of the evaluation. This finding suggests that the proposed transformation method is already helpful when the TAIs are not too complex and that by improving the interviewing method we can expect even better results.

With respect to the semantic evaluation, a slight tendency for better scores in the experimental group can be seen in general, however, only 2 of the measures for TAI 1 showed significantly better scores for the experimental group.

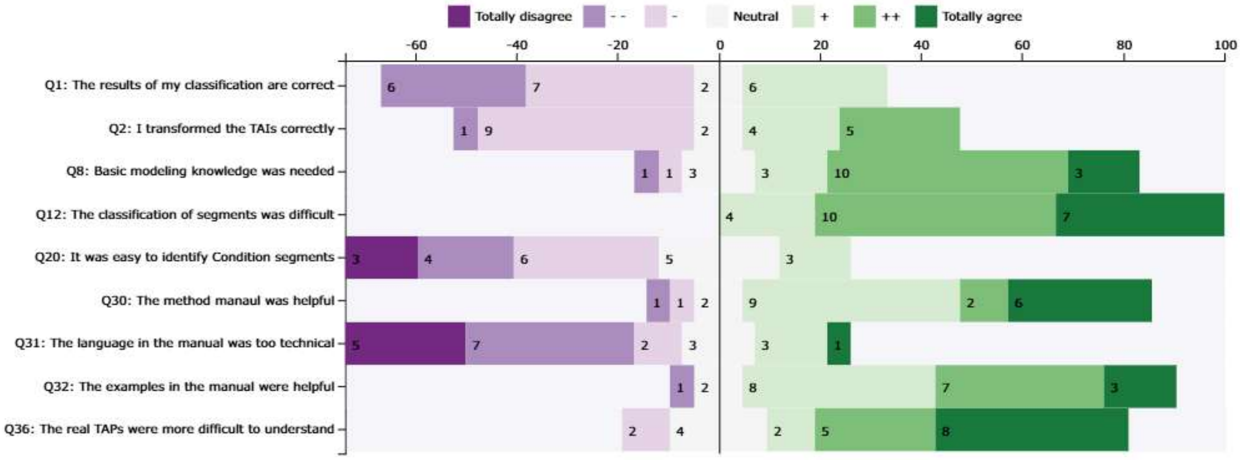

Figure 9. Histogram of answers to selected questions of the post-study questionnaire (Experimental group). Question text has been rephrased to fit the space.

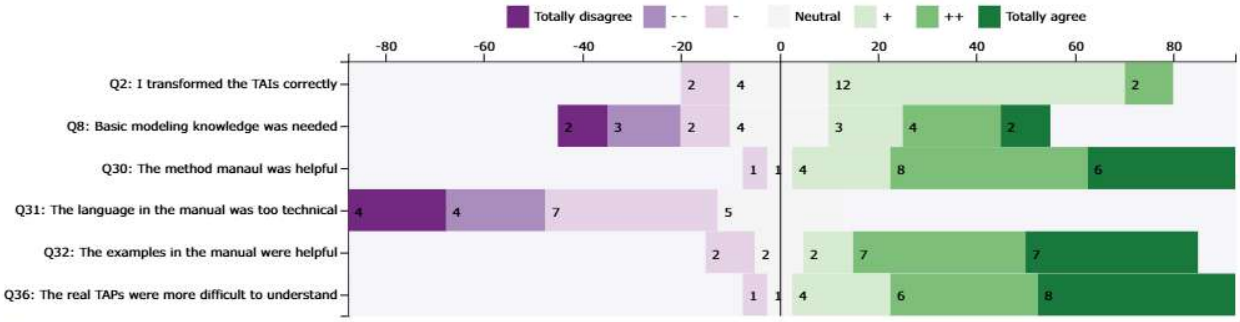

Figure 10. Histogram of answers to selected questions of the post-study questionnaire (Control Group). Question text has been rephrased to fit the space.

Furthermore, although the method did not address any of these measures directly, we were still able to observe some positive results, which further supports *Hypothesis 2*.

It is also interesting to note that participants in the experimental group felt less confident about their solutions, compared to the control group. This might be explained by the fact, that since the experimental group had more information about the transformation, they understood that correctly transforming the TAIs is not as trivial as it may appear.

6.1. Threats to validity

6.1.1. Statistical conclusion validity

For the power analysis, we used the partially post-hoc power analysis, since pure post-hoc power analyses (using the found effect sizes) simply re-express the found p-values. The partially post-hoc power analysis is superior since it provides information on how likely it was to observe a significant effect, given our sample, and given, in our case—medium effect size. We used a medium effect size, because this appeared us to be the most reasonable assumption given no prior research on our translation method exists and we neither wanted to be too conservative nor too lenient.

6.1.2. Internal validity

We were careful to provide the same information to both experimental and control group (except for the experimental material). This includes all instructions and input data for the tasks as well as the introduction to BPMN modeling; the control group even received the same pre-segmented texts that the experimental group used. As mentioned, the only difference in the material provided was related to the evaluated method: the reference to the method itself, a short tutorial with exercises and a post-study questionnaire about the method. Thus, we consider it safe to assume that differences found between the experimental and control group are due to the application of the proposed method.

6.1.3. Construct validity

We made sure that the experts who made the assessment of the participants' answers were completely blind to any information that could influence their evaluations. None of the experts had previous knowledge of the answers given by the participants before developing their own models. Also, they did not participate in the data-collection procedure and had no contact with any of the participants. They were also blind to the participants' group and identity. Both of the experts received a package with written instructions on how to develop their models and how to score the answers; including a template with a scoreboard that they could fill out. Additionally, the data-collection was completely performed by one person, who is one of the authors of this work. This is the only person who had direct contact with the participants, but only during the study. They did not participate as an expert in the assessment.

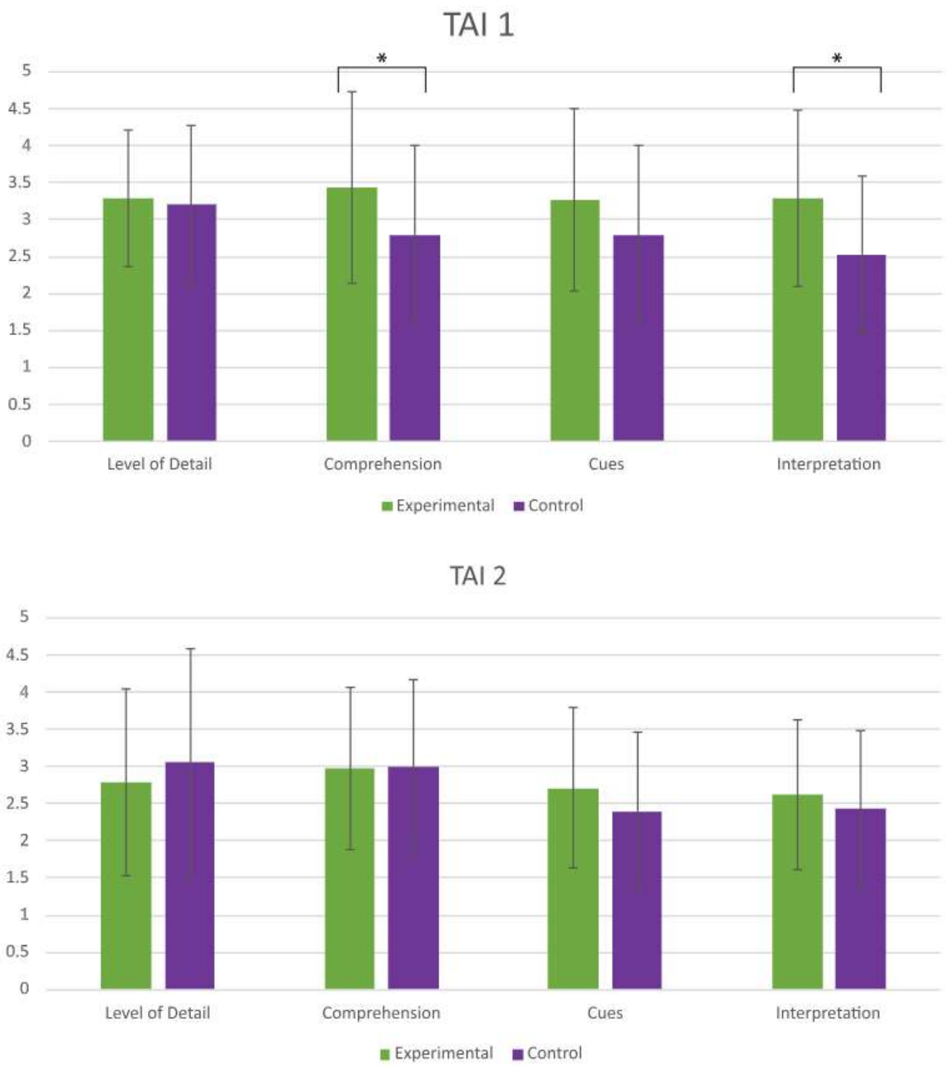

Figure 11. Results of the semantic evaluation per group for each TAI. Average scores were normalize to a scale of 1–5. (*) Indicate a significant effect of group.

6.1.4. External validity

To recruit the participants, we made use of a service and platform available at the university for such purpose. This platform has a database of possible participants with different profiles. Candidates that match the desired profile receive an email with a description of the study and link to register if they accept to participate. Mail reminders and any other communication with the participants is made through the platform and its administrators serve as intermediaries. None of the participants had any previous relation or even direct contact to the authors nor were involved with any of the work presented here. However, we note that the proposed transformation is fined-tuned to the German language. Some rules and examples might need to be adapted for use in other languages.

7. Conclusion and future work

In this work, we have outlined a transformation method to obtain BPMN models from verbal TAIs. The method is accompanied by the corresponding manual, which includes exercises and examples to help the modeler to better understand the method. We have tested the method and the manual in a study with an experimental and control group. In this study, participants had to perform transformations of two different TAIs, obtained from reports of real habit descriptions. The control group did this task with only an explanation to BPMN while the experimental group had additionally to use our proposed method. The goal was primarily to compare the results of the models between both groups to determine if the method influences the type of transformations that a modeler would choose. We also compared the solutions to expert models to check for accuracy of the solutions in both groups. The results of the study support our hypothesis that by applying the proposed transformation method we obtain solutions that are more similar to expert solutions. Our results suggest that using the transformation method reduces the influence of the individual modeler on the modeling result.

We are aware that using TAIs as a source material for habit descriptions might have made the transformation task unnecessarily difficult. However, we must consider that in the context of letting users of a potential persuasive system describe their habits as freely as possible and, thus, as close to their mental image of the habit in question as possible, we gain a great benefit due to a low influence of any structured method (they are not familiar with), which may hinder the end-users to expose their thinking. Furthermore, much of the context information provided in the TAIs (which made the task difficult) was needed to understand the habits.

We can also argue that by starting with this complex form of habit description, we have now a deeper understanding of the transformation method and habits in general. With this knowledge we can now design a more structured interview process to obtain habit descriptions from users without missing important information. With better interviews and thus better habit descriptions, using the method should help obtain more accurate models faster.

In addition, the method may have great potential to be applied to the creation of interactive systems of other types then persuasive systems. User-centered design processes are quasi standard in various industrial software developments and, thus, the presented approach may inspire future modeling and development processes in practice. For research, the method could be used to process qualitative data by simultaneously reducing the influence by the person who processes and thus reducing unwanted variance in the final data.

Therefore in future research, these potential applications need to be further investigated in the context of other system types, development processes in practice as well as research methods for handling qualitative data. Additional future research could also apply machine learning to automatize the transformations with the goal to support the manual transformation and to handle the complexity of the TAIs. Regardless, improving the way we obtain individualized habit models would allow us to design persuasive systems that are tailored to specific user needs, which support our goal to address habit change through the use of interactive systems.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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